

Overview & System Architecture

?

What is ESP-NOW

The robot is controlled by handheld ESP32 controllers over ESP-NOW, a connectionless, low-latency wireless protocol built into the ESP32 that sits below WiFi's usual AP/station overhead. There's no pairing handshake, no IP stack, and no dropped sessions to manage.

!

Why ESP-NOW (not WiFi/UDP or Bluetooth)

- ▶ ~1ms typical latency vs. tens of ms for a WiFi socket
- ▶ No access point required — works in the field or at a hackerspace
- ▶ Peer-to-peer: any ESP32 can talk to any other it knows the MAC of
- ▶ Packets map directly onto C structs — no JSON, no serialization

One Robot, Multiple Controllers

There is one robot, driven by multiple different controllers — it has no idea which one it's hearing from at any moment. The robot runs one fixed firmware image no matter which controller is talking to it; it just listens for the same packet types, from any peer, on the same MAC/channel.

Device	Role	MAC Address
Motor receiver (robot)	Drives motors, relay/lights	XX:XX:XX:XX:XX:XX
Relay receiver (robot)	Switches lights/accessories	XX:XX:XX:XX:XX:XX
Single-joystick controller	Sender — drive + lights	XX:XX:XX:XX:XX:XX
Dual-joystick controller	Sender + receiver — telemetry	XX:XX:XX:XX:XX:XX

Because the protocol is just two small, documented C structs sent as raw ESP-NOW packets, any bare ESP32 — flashed with nothing more than the struct definitions and the robot's MAC address — can control the robot. No authentication, no proprietary handshake.